Table R1: **Table 2 of the submission with the refined rule ($\delta = 0.3$).** Counterfactual Video Foley Generation on 11,890 VGGSound-Counterfactual triplets; NFE, the bold/underline ranking and †FAD (not ranked) as in the submission. Blue: new values of the refined rule (margin rule, $\delta = 0.3$). "$\alpha = 1.2$, submitted" rows: submitted values, black and not ranked. Baseline rows, incl. bold and underline marks, are exactly as submitted.

Method		Sound Identity Replacement					Temporal Alignment	Distribution /Quality		
		CLAP-T ↑	Clf. acc. ↑	FLAM-T ↑	ΔFLAM ↑	(+) Ratio ↑	DeSync ↓	IS ↑	PQ ↑	FAD†
(a) Comparison with existing VT2A systems										
Ours										
TAP-Foley	MMAudio	**0.299**	**0.901**	**0.960**	0.425	**0.916**	0.455	**21.16**	6.21	22.8
	Omni2Sound	0.242	0.833	0.891	0.301	0.824	0.501	19.98	6.30	24.7
TAP-Foley (α=1.2, submitted)	MMAudio	0.296	0.885	0.959	0.384	0.900	0.427	20.64	6.21	21.6
	Omni2Sound	0.241	0.829	0.887	0.299	0.821	0.510	19.83	6.31	24.5
Baseline										
ControlFoley		0.229	0.601	0.864	0.089	0.610	**0.316**	17.10	6.13	14.6
CAFA		0.243	0.786	0.942	0.194	0.785	0.685	12.21	6.36	22.3
FoleyCrafter		0.154	0.514	0.746	-0.004	0.478	1.244	10.74	6.06	30.4
ReWaS		0.005	0.443	0.344	-0.005	0.504	1.088	5.33	5.44	58.3
AudioX		0.187	0.475	0.840	-0.034	0.475	1.136	16.27	**6.53**	18.1
Baseline (with negative-source CFG)										
ControlFoley		0.255	**0.893**	0.910	**0.456**	**0.916**	0.754	15.66	5.89	23.6
CAFA		0.200	0.873	0.875	0.335	0.860	0.715	10.36	6.25	24.5
FoleyCrafter		0.105	0.732	0.662	0.281	0.742	1.264	8.08	6.02	40.8
ReWaS		-0.045	0.586	0.190	0.050	0.575	1.168	4.73	5.52	67.7
AudioX		0.217	0.845	0.873	0.329	0.853	1.233	14.79	6.19	25.2
(b) Comparison of inference-time guidance methods										
MMAudio										
TAP-Foley (ours)		**0.299**	0.901	**0.960**	0.425	0.916	0.455	**21.16**	**6.21**	22.8
TAP-Foley (α=1.2, submitted)		0.296	0.885	0.959	0.384	0.900	0.427	20.64	6.21	21.6
CounterFlow		0.292	**0.916**	0.949	**0.475**	**0.927**	0.649	19.75	6.16	23.7
Default CFG		0.162	0.350	0.727	-0.145	0.335	**0.323**	16.39	**6.21**	10.8
Negative-source CFG		0.212	0.758	0.824	0.280	0.778	0.703	12.93	5.93	19.2
Omni2Sound										
TAP-Foley (ours)		**0.242**	**0.833**	**0.891**	**0.301**	**0.824**	0.501	**19.98**	**6.30**	24.7
TAP-Foley (α=1.2, submitted)		0.241	0.829	0.887	0.299	0.821	0.510	19.83	6.31	24.5
CounterFlow		0.213	0.765	0.862	0.226	0.740	0.576	15.00	6.06	23.7
Default CFG		0.110	0.223	0.620	-0.267	0.195	**0.408**	18.58	6.27	14.2
Negative-source CFG		0.132	0.453	0.714	-0.029	0.438	0.642	11.35	6.05	27.1

Table R2: **Table 7 of the submission with the refined rule** ($\delta = 0.3$)**.** Onset-level evaluation on 1,280 Greatest Hits counterfactual triplets. Onset scores count only detected onsets with PE-A-Frame evidence for the target identity, so they measure identity and timing jointly; recall, precision and F1 use a 0.3-s tolerance, as in the submission. Panel (a) covers the two conflict-aware systems; panel (b) uses comparable NFE. Bold and underline mark the best and second-best values *across all rows in (a)* and *within each backbone in (b)*. Blue: new values of the refined rule (margin rule, $\delta = 0.3$). "$\alpha = 1.2$, submitted" rows: submitted values, black and not ranked. Baseline rows, incl. bold and underline marks, are exactly as submitted.

Method		Sound Identity Replacement		Onset Analysis		
		CLAP-T ↑	Clf. acc. ↑	Onset recall ↑	Onset precision ↑	Onset F1 ↑
(a) Comparison with existing VT2A systems						
Ours						
TAP-Foley	MMAudio	**0.347**	**0.997**	**0.864**	0.802	**0.797**
	Omni2Sound	0.303	0.952	0.809	0.775	0.757
TAP-Foley (α=1.2, submitted)	MMAudio	0.343	0.991	0.851	0.818	0.801
	Omni2Sound	0.297	0.948	0.803	0.780	0.758
Baseline						
ControlFoley		0.301	0.881	0.771	**0.833**	0.769
CAFA		0.305	0.956	0.775	0.802	0.756
Baseline (with negative-source CFG)						
ControlFoley		0.305	0.981	0.754	0.632	0.616
CAFA		0.276	0.976	0.767	0.722	0.704
(b) Comparison of inference-time guidance methods						
MMAudio						
TAP-Foley		**0.347**	**0.997**	**0.864**	0.802	**0.797**
TAP-Foley (α=1.2, submitted)		0.343	0.991	0.851	0.818	0.801
CounterFlow		0.305	0.981	0.705	0.673	0.631
Default CFG		0.295	0.877	0.746	**0.830**	0.751
Negative-source CFG		0.324	0.984	0.798	0.724	0.716
Omni2Sound						
TAP-Foley		**0.303**	**0.952**	**0.809**	**0.775**	**0.757**
TAP-Foley (α=1.2, submitted)		0.297	0.948	0.803	0.780	0.758
CounterFlow		0.249	0.913	0.713	0.632	0.612
Default CFG		0.174	0.541	0.495	0.574	0.506
Negative-source CFG		0.184	0.823	0.603	0.639	0.581

Table R3: **Family-stratified means on the full VGGSound-Counterfactual benchmark (refined rule, $\delta = 0.3$, matched NFE).** Mean ΔFLAM and DeSync of Default CFG, CounterFlow and **TAP-Foley** per source family, target family (seven families, App. C.1) and family relation (within-family: source and target in one family). In every stratum, **TAP-Foley** replaces more than Default CFG and has lower DeSync than CounterFlow; †: difference not significant (95% CI).

Stratum	Pairs	ΔFLAM ↑			DeSync ↓		
		Default CFG	CounterFlow	**TAP-Foley**	Default CFG	CounterFlow	**TAP-Foley**
MMAudio							
Full benchmark	11,890	-0.145	0.475	0.425	0.323	0.649	0.455
Source family							
Animals	1,890	-0.067	0.540	0.483	0.352	0.733	0.519
Human voice/vocal	880	-0.289	0.513	0.402	0.117	0.426	0.234
Instruments	4,490	-0.161	0.367	0.323	0.263	0.533	0.364
Other	3,340	-0.118	0.574	0.529	0.366	0.716	0.517
Sports	430	-0.143	0.330	0.322	0.328	0.695	0.441
Vehicles	650	-0.202	0.582	0.567	0.650	1.037	0.852
Water/weather	210	-0.171	0.402	0.311	0.481	0.936	0.582
Target family							
Animals	2,229	-0.132	0.557	0.477	0.291	0.661	0.458
Human voice/vocal	554	-0.324	0.518	0.456	0.154	0.401	0.258
Instruments	4,158	-0.108	0.358	0.313	0.233	0.509	0.332
Other	3,319	-0.147	0.522	0.492	0.428	0.786	0.570
Sports	707	-0.227	0.403	0.353	0.333	0.704	0.502
Vehicles	617	-0.206	0.683	0.633	0.512	0.875	0.678
Water/weather	306	-0.093	0.612	0.549	0.519	0.831	0.672
Family relation							
Within-family	8,060	-0.091	0.410	0.366	0.293	0.590	0.401
Cross-family	3,830	-0.260	0.610	0.550	0.384	0.771	0.568
Omni2Sound							
Full benchmark	11,890	-0.267	0.226	0.301	0.407	0.576	0.501
Source family							
Animals	1,890	-0.264	0.306	0.387	0.476	0.695	0.593
Human voice/vocal	880	-0.401	0.292	0.428	0.226	0.365	0.319
Instruments	4,490	-0.264	0.151	0.233	0.313	0.438	0.406
Other	3,340	-0.232	0.269	0.326	0.463	0.663	0.556
Sports	430	-0.135	0.091	0.162	0.409	0.509	0.482†
Vehicles	650	-0.349	0.306	0.336	0.731	0.905	0.754
Water/weather	210	-0.393	0.186	0.218	0.680	1.079	0.874
Target family							
Animals	2,229	-0.254	0.360	0.432	0.406	0.632	0.549
Human voice/vocal	554	-0.406	0.286	0.432	0.263	0.370	0.319
Instruments	4,158	-0.216	0.147	0.221	0.295	0.409	0.371
Other	3,319	-0.328	0.198	0.261	0.522	0.727	0.610
Sports	707	-0.247	0.182	0.257	0.389	0.486	0.472†
Vehicles	617	-0.295	0.287	0.402	0.586	0.790	0.671
Water/weather	306	-0.136	0.498	0.517	0.652	0.953	0.806
Family relation							
Within-family	8,060	-0.225	0.175	0.244	0.377	0.530	0.456
Cross-family	3,830	-0.356	0.334	0.421	0.472	0.674	0.598